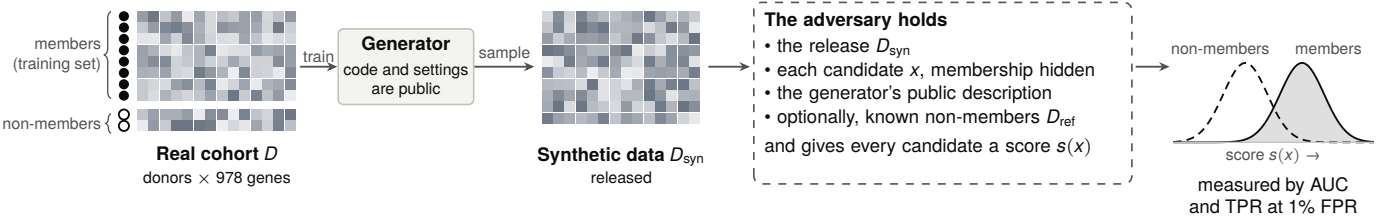
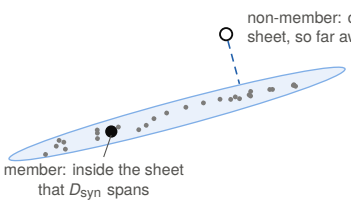
a The membership game



b Three attacks, each reading the place where a generator family keeps its training donors

MahalaMIA — reads the covariance

against MVN and the CVAE

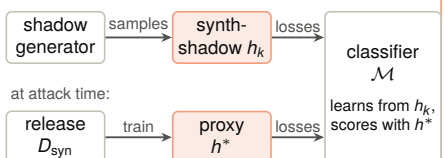


$s(x)$ is minus the Mahalanobis distance from x to D_{syn}. No shadow models; seconds to run.

MeLoMIA — reads the losses

against CVAE, NoisyDiffusion, TabSyn, TabPFN

repeated K times, on known splits of D:

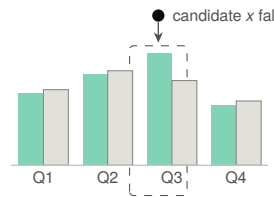


trained on synthetic data only, so the classifier sees the same kind of model twice

Each loss is calibrated against the same profile's losses under the K synth-shadows.

MAMA-MIA — reads the measured marginals

against the DP graphical models



$s(x) = \sum_t \log \hat{p}^t_{\text{syn}}(x_t) \,/\, \hat{p}^t_{\text{ref}}(x_t)$ over the measured tables t: a member added one count to its cell of each, underneath the DP noise.